Lab 3.3
SPC Control Chart and Capability

Goal	Build an Xbar-R control chart from baseline data and read the monitoring signal.
You will produce	Xbar and R per subgroup, frozen baseline limits, a plotted chart, and Cp/Cpk with a caution.
Time box	30 minutes
Outcome	LO1 explain SPC · LO2 build the chart with Copilot in Excel
Case	CW-401 chemical dosing skid

All records are fictional training data for classroom use. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.

Before you start

Open the Starter workbook for this lab. Do not open Worked-Reference.xlsx until you have finished your own attempt and checked it. Sign in to Microsoft 365 at https://m365.cloud.microsoft/ with your training account.

The concept in one minute

SPC separates the noise a process always has from a change worth investigating. Common cause variation is the process being itself. Special cause variation is something new. Control limits come from the process, calculated on a stable baseline and then frozen, so later points are judged against how the process behaved when it was healthy. Control limits are not specification limits: one describes what the process does, the other what the customer requires. A point outside the limits is a signal to investigate, never a diagnosis.

Step-by-step

Step 1 — Open the workbook and read the structure

Open **Starter.xlsx** and look at **SPC_Data**.

There are 16 subgroups of four measurements each. Subgroups 1 to 12 are Baseline; 13 to 16 are Monitoring. The Xbar and R columns are empty.

Step 2 — Calculate Xbar and R for one subgroup by hand first

Before using Copilot, do subgroup 1 yourself so you know what right looks like:

```
= AVERAGE(61.8, 62.4, 61.5, 62.1) = 61.950
= MAX - MIN = 62.4 - 61.5 = 0.900
```

This takes thirty seconds and it is the reason you will catch an error later.

Step 3 — Open Copilot and paste the prompt

Select inside **SPCData**, open Copilot in Excel and paste the prompt from **Lab-3.3-Prompt.pdf**.

The prompt supplies the constants for n = 4: A2 = 0.729, D4 = 2.282 and d2 = 2.059. Copilot should not be guessing these.

Step 4 — Check that limits came from the BASELINE only

This is the step most people get wrong. The limits must be calculated from subgroups 1 to 12 and then held fixed.

If Copilot used all 16 subgroups, the drifting monitoring points drag the limits down with them and the signal disappears. Ask it to recalculate on the baseline only.

Expected: Xbarbar approximately 62.027, Rbar approximately 0.658.

```
= UCL = 62.027 + 0.729 x 0.658 = 62.507
= LCL = 62.027 - 0.729 x 0.658 = 61.547
```

Step 5 — Plot all 16 subgroups against the frozen limits

Ask Copilot for a line chart of Xbar for all 16 subgroups with the centre line, UCL and LCL drawn as horizontal reference lines.

The baseline points should sit inside the limits without a pattern. The monitoring points should fall away below the LCL.

Step 6 — Read the signal correctly

Check subgroup 13 by hand:

```
= AVERAGE(61.5, 61.2, 61.4, 61.0) = 61.275, which is below the LCL of 61.547
```

All four monitoring subgroups are below the LCL and they trend steadily downwards rather than jumping once. A drift like this suggests something progressive, such as wear or fouling, rather than a single event.

Write the conclusion as a signal to investigate. Do not write a cause.

Step 7 — Calculate capability and read it with caution

Ask for sigma, Cp and Cpk. Expect sigma approximately 0.320, Cp approximately 3.13 and Cpk approximately 1.01.

Cp is three times Cpk. That gap does not mean the process cannot hold the tolerance; it means the process is running off centre, high in the band.

State the larger caution too: capability is only meaningful while the process is stable, and these monitoring subgroups show it is not. Reporting Cpk alone here would mislead.

Step 8 — Compare with the worked reference

Open **Worked-Reference.xlsx** and confirm your limits, sigma, Cp and Cpk match.

Check your own work

Do these by hand before you accept the Copilot output. If a number disagrees, the Copilot answer is wrong until proven otherwise.

- Subgroup 1 Xbar by hand: (61.8+62.4+61.5+62.1)/4 = 61.950, R = 0.900.
- Limits were calculated on subgroups 1-12 only and then frozen.
- Subgroup 13 Xbar = 61.275, below the LCL of 61.547.
- All four monitoring subgroups fall below the LCL and trend downwards.
- Cp much greater than Cpk was explained as off-centre, not incapable.
- The output says capability is not meaningful while the process is unstable.

Engineering boundary

A point outside a control limit is a signal to investigate, never a diagnosis and never proof that equipment is unsafe. Do not recalculate limits using the monitoring data. Do not report capability as though the process were stable. Root cause requires physical investigation by a competent engineer.

Evidence to retain

Save the final prompt, the workbook, one independent hand check, any correction you made and the name of the human reviewer.

All records are fictional training data for classroom use. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.